

ComicCraft: AI Comic Story Creator using Gemini Models

Team ID: SWTID-2026-5650 | SmartBridge AI & GenAI Project Track

Project Description:

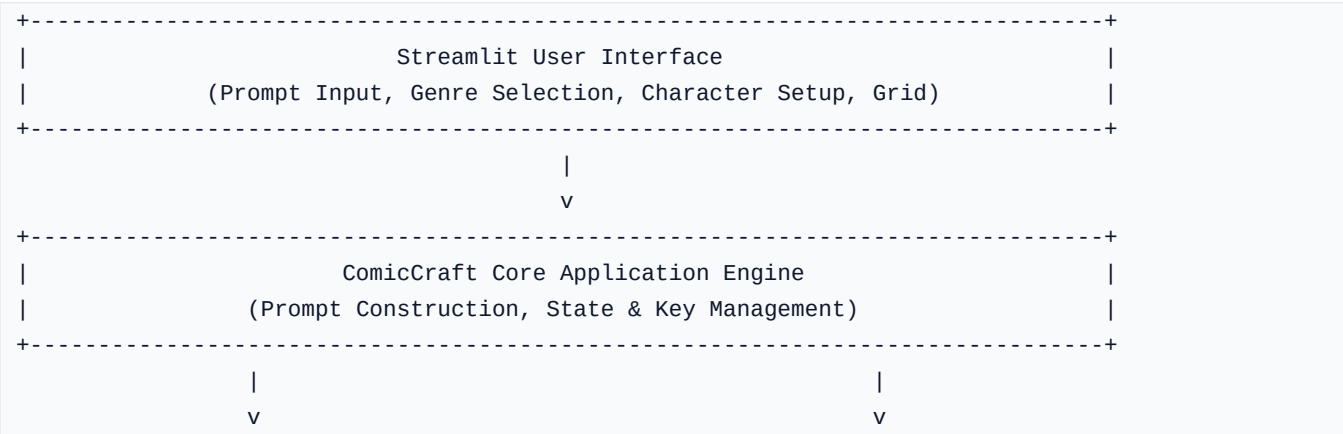
ComicCraft harnesses Generative AI to provide an intelligent, automated comic story creation experience. The platform integrates a Script & Dialogue Generator module, which produces multi-panel comic scripts, visual scene descriptions, character dialogue, and sound effects, alongside a Comic Art Generator that synthesizes vibrant, panel-by-panel comic illustrations matching the generated storyline.

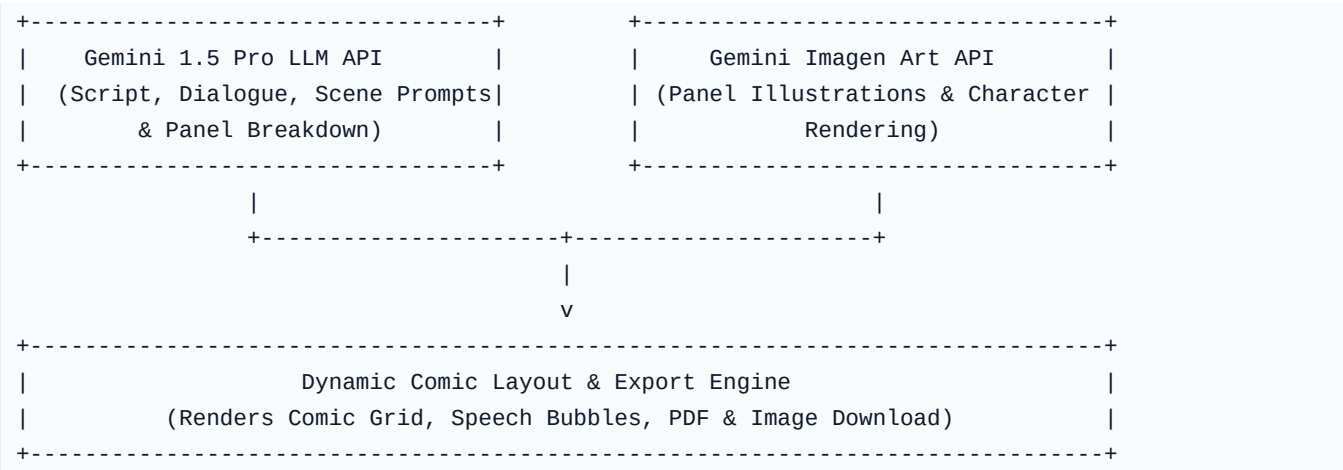
Utilizing Google's Gemini API with the Gemini 1.5 Pro and Gemini Imagen models, ComicCraft processes user prompts (genre, theme, plot outline, and character details) to deliver rich, custom graphic narratives. Built with Python and Streamlit and deployed publicly, the platform ensures seamless user interaction, secure API key handling via .env, and dynamic comic layout export options.

Scenarios

- Scenario 1:** A graphic novelist inputs a prompt: *'A sci-fi detective in a neon-lit cyberpunk city investigating a mysterious glitch.'* ComicCraft generates a 4-panel noir comic script with moody dialogue, renders 4 consistent visual panels, and formats them into a polished comic grid with speech bubbles.
- Scenario 2:** An educator requests a comic explaining *'The journey of electricity through a circuit.'* The system generates an engaging, 3-panel educational comic strip featuring fun character personifications and informative dialogue bubbles.
- Scenario 3:** A social media creator selects a comedy theme: *'A mischievous puppy trying to hide a slippers.'* The AI produces an expressive, 3-panel humor strip with punchy dialogue and high-vibrancy artwork optimized for sharing.

Technical Architecture:





Description: ComicCraft utilizes a modular tier architecture to deliver rapid, AI-driven comic generation. The frontend provides an interactive UI built with Streamlit. The backend orchestrates prompt engineering and manages Google Gemini API integration using Gemini 1.5 Pro for script composition and Gemini Imagen for panel rendering, securing access tokens locally via .env.

Pre-requisites:

- **Streamlit Framework Knowledge:** [Streamlit Documentation](#)
- **Google Gemini API / GenAI SDK Familiarity:** [Google AI Studio Documentation](#)
- **HTML, CSS, and UI Layout Skills:** [W3Schools HTML/CSS Tutorials](#)
- **Python Programming Proficiency:** [Python Documentation](#)
- **Version Control with Git:** [Git Documentation](#)
- **Development Environment Setup:** [Python Virtual Environment Guide](#)
- **Public Deployment Setup:** [Streamlit Cloud Deployment Guide](#)

Project Workflow:

Milestone 1: Model Selection and Architecture

- Activity 1.1: Generate a Gemini API key and configure it securely in the project environment.
- Activity 1.2: Research and select appropriate Generative AI models (Gemini 1.5 Pro for scriptwriting, Gemini Imagen for comic art).
- Activity 1.3: Define application architecture, detailing interactions between frontend, story engine, image generator, and API integration.
- Activity 1.4: Set up development environment, installing necessary libraries (google-generativeai, streamlit, pillow, python-dotenv).

Milestone 2: Core Functionalities Development

- Activity 2.1: Develop core functionalities: Comic Script Generator, Character Prompt Anchoring, and Speech Bubble Renderer.
- Activity 2.2: Implement backend logic to manage routing and prompt construction, ensuring smooth API interactions.

Milestone 3: App.py Development

- Activity 3.1: Write main application logic in app.py, establishing interface layouts, state management, and integrating Gemini models.

Milestone 4: Frontend Development

- Activity 4.1: Design and develop user interface using Streamlit, ensuring a responsive comic panel layout.
- Activity 4.2: Create dynamic preview templates to render multi-panel comics with overlay text.

Milestone 5: Deployment

- Activity 5.1: Prepare application for local deployment by configuring server environment and dependencies.
- Activity 5.2: Deploy application publicly over a secure URL using Streamlit Community Cloud.

Milestone 6: Conclusion

- Activity 6.1: Evaluate project performance, document user feedback, and compile final documentation.

Milestone 1 Detail: Model Selection & Architecture

In this milestone, we focus on selecting appropriate generative AI models from Google Gemini for our comic creation needs. This involves researching model capabilities, ensuring the chosen models align with generating structured comic scripts, dialogue, visual descriptions, and panel artwork.

Activity 1.1: Generate a Gemini API Key

Before the application can communicate with Google Gemini models, you need a valid Gemini API key. Follow these steps:

- **Step 1 — Create a Google AI Studio Account:** Visit <https://aistudio.google.com/> and sign up using your Google account.
- **Step 2 — Navigate to API Keys:** After logging in, click on the profile/settings icon or select **API Keys** from the navigation menu.
- **Step 3 — Create a New API Key:** Click the 'Create API Key' button. Give your key a descriptive name (e.g., comiccraft-api-key) so it is easy to identify.
- **Step 4 — Copy API Key:** Once generated, copy the API key immediately and store it securely.
- **Step 5 — Add the Key to Your Project:** Create a .env file in the root directory of your project and paste the key as shown below:
GEMINI_API_KEY=your_copied_api_key_here

- **Step 6 — Verify Key is Loaded:** The app.py file loads this key automatically at startup using python-dotenv:
from dotenv import load_dotenv
import os, google.generativeai as genai
load_dotenv()
genai.configure(api_key=os.environ.get('GEMINI_API_KEY'))

- **Important Security Rule:** Never commit your .env file to a public GitHub repository. Always add .env to your .gitignore file.